

"PAPER_{0:D3}" -f [int]# PAPER #93 ■ M87* Event Horizon: UQFF Field Analysis

Author: Daniel T. Murphy

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Title: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ([SCm] ■ 0.99, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** validateuqffmuge.py, from_system('M87'), EHT 2019-2024 data **Index Slot:** ■1.12 UQFF Master Calculators, \$n = [int]# PAPER #93 ■ M87* Event Horizon: UQFF Field Analysis

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Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ([SCm] ■ 0.99, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** validateuqffmuge.py, fromsystem('M87'), EHT 2019-2024 data **Index Slot:** ■1.12 UQFF Master Calculators, PAPER093

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \text{ } \blacksquare \text{ } 10^? \text{ } M?$, $d = 16.8 \text{ Mpc}$), provides a strong-field test of UQFF at a mass $1,600 \text{ } \blacksquare \text{ } \text{Sgr A}$. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{UQFF} = 0.99 \text{ TH} = 1.34 \text{ } \blacksquare \text{ } 10^? \text{ } \blacksquare \text{ } 7 \text{ K}$. All 8 terms are finite and `g_total` is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \text{ } \blacksquare \text{ } 10^? \text{ } 4 \text{ day?} \blacksquare$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. M87* System Parameters

Parameter	Value	Source
M_BH	$1.26 \text{ } \blacksquare \text{ } 104 \text{ } \blacksquare \text{ kg}$ ($6.5 \text{ } \blacksquare \text{ } 10^? \text{ } M?$)	EHT 2019 (first image)
r_Schwarzschild	$1.92 \text{ } \blacksquare \text{ } 10 \text{ } \blacksquare \text{ } \blacksquare \text{ m}$	$2GM/c \blacksquare$
r_horizon (UQFF)	$1.95 \text{ } \blacksquare \text{ } 10 \text{ } \blacksquare \text{ } \blacksquare \text{ m}$	$r_S \text{ } \blacksquare \text{ } (1 + 0.015)$
Distance	$16.8 \text{ Mpc} = 5.18 \text{ } \blacksquare \text{ } 10 \text{ } \blacksquare \text{ } \blacksquare \text{ m}$	Virgo Cluster
Spin (a/M)	$0.90 \text{ } \blacksquare \text{ } 0.05$	EHT 2024
Jet power P_jet	$\sim 1044 \text{ erg/s}$	VLA/VLBI
T_H (GR)	$1.35 \text{ } \blacksquare \text{ } 10^? \text{ } \blacksquare \text{ } 7 \text{ K}$	$?c \blacksquare / (8pGMk_B)$
T_UQFF	$1.34 \text{ } \blacksquare \text{ } 10^? \text{ } \blacksquare \text{ } 7 \text{ K}$	$T_H \text{ } \blacksquare \text{ } 0.99$

2. 8-Term MUGE at $r_{\text{horizon}} = 1.95 \times 10^{17}$ m

Term	Value (m/s)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	Ug4 ? M/r^6 ? M87 larger r offsets large M
U_i	0.14	
cosmological	-9.1×10^{-10}	? negligible at horizon
quantum	$+2.0 \times 10^{-4}$	Planck-scale
fluid	$+6.2 \times 10^{-10}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, >> far_field
g_total	2211	100%

Newtonian $g_{\text{Newton}} = 2207$ m/s. MUGE total = 2211 m/s ? UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $\Gamma \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With $\eta = \eta_{\text{acc}}/\eta_{\text{Edd}} \approx 10^{-3}$ (low state) and $[\text{SCm}] = 0.99$:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $\eta_{\text{jet}} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $\theta_{\text{ring}} = 42 \pm 3 \mu\text{as}$? *physical* $r_{\text{ring}} = 5.0 \text{ GM}/c$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via $[\text{SCm}]$:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [\text{SCm}]}{2}} = r_{\text{GR}} \times \sqrt{1.005} \approx 1.0025 \cdot r_{\text{GR}}$$

$\Delta \theta = +0.25\% \approx 0.1 \mu\text{as}$ shift (undetectable by current EHT at $\pm 3 \mu\text{as}$ precision).

5. Coherence vs Distance

At M87 with its much larger r_{horizon} (vs Sgr A):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon (1.95×10^{17} m)	1.0	$g_{\text{coh},0}$	1.000
1 kpc (3.1×10^{17} m)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: assert coh_at_horizon > coh_far * 1e6 **■ PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_{\rm H}^{\rm M87*} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\rm UQFF}^{\rm M87*} = 0.9999 \times T_{\rm H} = 1.34 \times 10^{-17} \text{ K}$$

? = 0.01% reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	? PASS
g_total = Newton + 0.18%	? PASS
No NaN/Inf for M87*	? PASS
Coherence peak at horizon	? PASS
Jet power UQFF estimate	3.6■1044 erg/s (consistent)
Shadow diameter deviation	0.25% (■ EHT precision)
T_UQFF	1.34 ■ 10?■7 K

Source: validateuqffmuge.py | fromsystem('M87') | EHT 2019-2024 | [SCm]=0.99 | all 8 terms PASS
.Groups[1].Value "PAPER{0:D3}" -f \$n

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_\odot$, $d = 16.8 \text{ Mpc}$), provides a strong-field test of UQFF at a mass 1,600■ Sgr A. The from_system('M87') constructor in validate_uqff_muge.py encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{\rm UQFF} = 0.99 T_{\rm H} = 1.34 \times 10^{-17} \text{ K}$. All 8 terms are finite and g_total is consistent with the VLBI ring diameter.

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1. M87* System Parameters

Parameter	Value	Source
M_BH	1.26 ■ 104■ kg (6.5■10? M?)	EHT 2019 (first image)
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Parameter	Value	Source
Distance	16.8 Mpc = 5.18×10^{24} m	Virgo Cluster
Spin (a/M)	0.90 ± 0.05	EHT 2024
Jet power P_{jet}	~ 1044 erg/s	VLA/VLBI
T_{H} (GR)	1.35×10^7 K	$c^2/(8\pi G M_{\text{K}} B)$
T_{UQFF}	1.34×10^7 K	$T_{\text{H}} \times 0.99$

2. 8-Term MUGE at $r_{\text{horizon}} = 1.95 \times 10^{24}$ m

Term	Value (m/s)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	Ug4 ? M/r^6 ? M87 larger r offsets large M
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$\Delta r \approx +0.25\% \approx 0.1 \text{ as}$ shift (undetectable by current EHT at $\pm 3 \text{ as}$ precision).

5. Coherence vs Distance

At M87 with its much larger r_{horizon} (vs Sgr A):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon (1.95×10^{17} m)	1.0	$g_{\text{coh},0}$	1.000
1 kpc (3.1×10^{22} m)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_{\text{H}}^{\text{M87*}} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\text{UQFF}}^{\text{M87*}} = 0.9999 \times T_{\text{H}} = 1.34 \times 10^{-17} \text{ K}$$

$\approx 0.01\%$ reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	? PASS
$g_{\text{total}} = \text{Newton} + 0.18\%$	? PASS
No NaN/Inf for M87*	? PASS
Coherence peak at horizon	? PASS
Jet power UQFF estimate	3.6×10^{44} erg/s (consistent)
Shadow diameter deviation	0.25% (EHT precision)
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Source: `validateuqffmuge.py` | `from_system('M87')` | EHT 2019-2024 | $[SC_m]=0.99$ | all 8 terms PASS
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Jet power P_jet	$\sim 1044 \text{ erg/s}$	VLA/VLBI
T_H (GR)	$1.35 \times 10^7 \text{ K}$	$\hbar c^3 / (8\pi G M k_B)$
T_UQFF	$1.34 \times 10^7 \text{ K}$	$T_H \times 0.99$

2. 8-Term MUGE at r_horizon = $1.95 \times 10^6 \text{ m}$

Term	Value (m/s ²)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	$Ug_4 \propto M/r^6$? M87 larger r offsets large M
U_i	0.14	
cosmological	-9.1×10^{-10}	? negligible at horizon
quantum	$+2.0 \times 10^{-4}$	Planck-scale
fluid	$+6.2 \times 10^{-10}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, >> far_field
g_total	2211	100%

Newtonian $g_{\text{Newton}} = 2207 \text{ m/s}^2$. MUGE total = 2211 m/s² ? UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $\Gamma \approx 6$) is mediated by Ug3 string rotation in the UQFF:

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$\Delta r \approx +0.25\% \approx 0.1 \mu\text{as}$ shift (undetectable by current EHT at $\pm 3 \mu\text{as}$ precision).

5. Coherence vs Distance

At M87 with its much larger r_{horizon} (vs Sgr A):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon ($1.95 \times 10^7 \text{ m}$)	1.0	$g_{\text{coh},0}$	1.000
1 kpc ($3.1 \times 10^{16} \text{ m}$)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_{\text{H}}^{\text{M87*}} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

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$\eta = 0.01\%$ reduction from GR. IceCube and FRB backgrounds: consistent.

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All 8 MUGE terms finite	? PASS
g_total = Newton + 0.18%	? PASS
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Location	$r/r_{\rm horizon}$	$g_{\rm coh}$	Ratio
At horizon (1.95×10 ¹⁰ m)	1.0	$g_{\rm coh,0}$	1.000
1 kpc (3.1×10 ¹⁶ m)	1.6×10 ⁶	~0	~10 ⁻⁶

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

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Title: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($[SCm]$ ■ 0.99, $[SSq] = 0.57$) **Date:** March 7, 2026 **Source Data:** `validateuqffmuge.py`, `fromsystem('M87')`, *EHT 2019-2024 data* **Index Slot:** ■1.12 *UQFF Master Calculators, PAPER093*

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_{\odot}$, $d = 16.8$ Mpc), provides a strong-field test of UQFF at a mass 1,600■ Sgr A. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{UQFF} = 0.99 T_{\rm H} = 1.34 \times 10^{-17}$ K. All 8 terms are finite and `g_total` is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day τ , $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. M87* System Parameters

Parameter	Value	Source
M_BH	$1.26 \times 10^4 \text{ kg}$ ($6.5 \times 10^7 \text{ M?}$)	EHT 2019 (first image)
r_Schwarzschild	$1.92 \times 10^{10} \text{ m}$	$2GM/c^2$
r_horizon (UQFF)	$1.95 \times 10^{10} \text{ m}$	$r_S (1 + 0.015)$
Distance	$16.8 \text{ Mpc} = 5.18 \times 10^{24} \text{ m}$	Virgo Cluster
Spin (a/M)	0.90 ± 0.05	EHT 2024
Jet power P_jet	$\sim 1044 \text{ erg/s}$	VLA/VLBI
T_H (GR)	$1.35 \times 10^7 \text{ K}$	$\tau c^2 / (8\pi G M k_B)$
T_UQFF	$1.34 \times 10^7 \text{ K}$	$T_H \times 0.99$

2. 8-Term MUGE at r_horizon = $1.95 \times 10^{10} \text{ m}$

Term	Value (m/s τ)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	$Ug_4 \times M/r_6 \times M87$ larger r offsets large M
U_i	0.14	
cosmological	-9.1×10^{10}	τ negligible at horizon
quantum	$+2.0 \times 10^{24}$	Planck-scale
fluid	$+6.2 \times 10^{10}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, $\gg$ far_field
g_total	2211	100%

Newtonian $g_{\text{Newton}} = 2207 \text{ m/s}^2$. MUGE total = 2211 m/s^2 τ UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $G \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With $\tau = \tau_{\text{acc}}/\tau_{\text{Edd}} \approx 10^{-3}$ (low state) and $[\text{SCm}] = 0.99$:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $\tau_{\text{jet}} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $r_{\text{ring}} = 42 \pm 3 \text{ } \mu\text{as}$ $r_{\text{physical ring}} = 5.0 \text{ GM}/c$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via $[SC_m]$:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [SC_m]}{2}} = r_{\text{GR}} \cdot \sqrt{1.005} \approx 1.0025 \cdot r_{\text{GR}}$$

$\Delta r = +0.25\% \pm 0.1 \text{ } \mu\text{as}$ shift (undetectable by current EHT at $3 \text{ } \mu\text{as}$ precision).

5. Coherence vs Distance

At M87 with its much larger r_{horizon} (vs Sgr A):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon ($1.95 \times 10^7 \text{ m}$)	1.0	$g_{\text{coh},0}$	1.000
1 kpc ($3.1 \times 10^{16} \text{ m}$)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_{\text{H}}^{\text{M87*}} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\text{UQFF}}^{\text{M87*}} = 0.9999 \times T_{\text{H}} = 1.34 \times 10^{-17} \text{ K}$$

$\Delta T = 0.01\%$ reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	? PASS
$g_{\text{total}} = \text{Newton} + 0.18\%$	? PASS
No NaN/Inf for M87*	? PASS
Coherence peak at horizon	? PASS
Jet power UQFF estimate	$3.6 \times 10^{44} \text{ erg/s}$ (consistent)
Shadow diameter deviation	0.25% ($\pm$ EHT precision)
T_{UQFF}	$1.34 \times 10^{-17} \text{ K}$

Source: `validateuqffmuge.py` | `fromsystem('M87')` | EHT 2019-2024 | $[SC_m]=0.99$ | all 8 terms PASS
Groups[1].Value "PAPER{0:D3}" -f \$n

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_\odot$, $d = 16.8$ Mpc), provides a strong-field test of UQFF at a mass 1,600 $\times$ Sgr A. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{UQFF} = 0.99 T_H = 1.34 \times 10^7$ K. All 8 terms are finite and `g_total` is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis $\times$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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Jet power P_jet	~ 1044 erg/s	VLA/VLBI
T_H (GR)	1.35×10^7 K	$\tau c^2 / (8\pi G M k_B)$
T_UQFF	1.34×10^7 K	$T_H \times 0.99$

2. 8-Term MUGE at r_horizon = 1.95×10^6 m

Term	Value (m/s 2)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	$Ug_4 \times M / r^6 \times M87$ larger r offsets large M
U_i	0.14	
cosmological	-9.1×10^4	$\times$ negligible at horizon
quantum	$+2.0 \times 10^4$	Planck-scale
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coherence	peaked at horizon	Gaussian, $\gg$ far_field
g_total	2211	100%

Newtonian $g_{\text{Newton}} = 2207$ m/s 2 . MUGE total = 2211 m/s 2 $\times$ UQFF excess = +0.18%.

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The M87 jet (1.4 kpc visible, Lorentz factor $\Gamma \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\rm jet}^{\rm UQFF} = U_{g3}(r_{\rm ISCO}) \cdot \dot{M}_{\rm acc} c^2 \cdot [\rm SCm]$$

With $\eta = \eta_{\rm acc}/\eta_{\rm Edd} \approx 10^{-3}$ (low state) and $[\rm SCm] = 0.99$:

$$P_{\rm jet}^{\rm UQFF} = 0.99 \times 10^{-3} L_{\rm Edd} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $\eta_{\rm jet} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $\theta_{\rm ring} = 42 \pm 3 \mu\text{as}$ *vs* $\theta_{\rm physical ring} = 5.0 \text{ GM}/c$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via $[\rm SCm]$:

$$r_{\rm shadow}^{\rm UQFF} = r_{\rm shadow}^{\rm GR} \cdot \sqrt{1 + \frac{1 - [\rm SCm]}{2}} = r_{\rm GR} \cdot \sqrt{1.005} \approx 1.0025 r_{\rm GR}$$

$\Delta \theta \approx +0.25\% \approx 0.1 \mu\text{as}$ shift (undetectable by current EHT at $3 \mu\text{as}$ precision).

5. Coherence vs Distance

At M87 with its much larger $r_{\rm horizon}$ (vs Sgr A):

Location	$r/r_{\rm horizon}$	$g_{\rm coh}$	Ratio
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From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

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$$T_{\rm H}^{\rm M87*} = \frac{\hbar c^3}{8\pi G M k_{\rm B}} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

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$$T_{\rm UQFF}^{\rm M87*} = 0.9999 \times T_{\rm H} = 1.34 \times 10^{-17} \text{ K}$$

$\eta = 0.01\%$ reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	η PASS
$g_{\rm total} = \text{Newton} + 0.18\%$	η PASS
No NaN/Inf for M87*	η PASS

Validation	Result
Coherence peak at horizon	? PASS
Jet power UQFF estimate	3.6■1044 erg/s (consistent)
Shadow diameter deviation	0.25% (■ EHT precision)
T_UQFF	1.34 ■ 10?■7 K

Source: validateuqffmuge.py | from_system('M87') | EHT 2019-2024 | [SCm]=0.99 | all 8 terms PASS
.Groups[1].Value

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M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \text{ } \blacksquare 10? \text{ } M?$, $d = 16.8 \text{ Mpc}$), provides a strong-field test of UQFF at a mass 1,600■ Sgr A. The from_system('M87') constructor in validate_uqff_muge.py encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $TUQFF = 0.99 \text{ TH} = 1.34 \text{ } \blacksquare 10? \blacksquare 7 \text{ K}$. All 8 terms are finite and g_total is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \blacksquare 10?4 \text{ day?} \blacksquare$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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Jet power P_jet	~1044 erg/s	VLA/VLBI
T_H (GR)	1.35 ■ 10?■7 K	?c■/(8pGMk_B)
T_UQFF	1.34 ■ 10?■7 K	T_H ■ 0.99

2. 8-Term MUGE at r_horizon = 1.95 ■ 10■■■ m

Term	Value (m/s■)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	Ug4 ? M■/r6 ? M87 larger r offsets large M■
U_i	0.14	
cosmological	-9.1 ■ 10?■■■	? negligible at horizon
quantum	+2.0 ■ 10?4■	Planck-scale
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Term	Value (m/s)	Notes
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Jet power UQFF estimate	3.6×10^{44} erg/s (consistent)
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$T_{\rm H}$ (GR)	1.35×10^{-17} K	$?c^2/(8\pi G M k_B)$
$T_{\rm UQFF}$	1.34×10^{-17} K	$T_{\rm H} \times 0.99$

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sum_Ug	3.75	Ug4 ? M ² /r6 ? M87 larger r offsets large M ²
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cosmological	-9.1 × 10 ⁻¹⁰	? negligible at horizon
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At M87 with its much larger $r_{\rm horizon}$ (vs Sgr A):

Location	r/r_horizon	g_coh	Ratio
At horizon (1.95×10 ¹⁰ m)	1.0	g_coh,0	1.000
1 kpc (3.1×10 ¹⁶ m)	1.6×10 ⁶	~0	~10 ⁻⁶

From validator: assert coh_at_horizon > coh_far * 1e6 **PASS** for M87* system.

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UQFF computed: Canonical UQFF buoyancy parameter $U_{\rm bi} = ?■[SSq]■GM/r■ = 5.0e-4■0.57■6.67e-11■M/r■$; for solar parameters: $U_{\rm bi,Sun} = 5.7e-4■6.67e-11■1.99e30/(6.96e8)■ = 1.47e+2 \text{ m/s}■$.